From Pairwise Preferences to Confidence Bounds

Contextual Dueling Bandits

Sections 1–4 · About 10 minutes · v2

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Compare two arms; observe one preference

Contextual: arms have features, and the decision set may change each round.

1 Observe the arm set

$$\mathcal{X}_t \subseteq \mathbb{R}^d$$

Each arm has features x.

2 Select a pair

$$x_{t,1},\ x_{t,2} \in \mathcal{X}_t$$

Example: two recommendations.

3 Observe a preference

$$y_t \in \{0, 1\}$$

y = 1: the first arm wins.

Latent utility $f(x) = \langle \theta^\star, x \rangle$

Difference feature $\tilde{x}_t = x_{t,1} - x_{t,2}$

Source: Section 1, p.1; Eqs. (1)–(3)

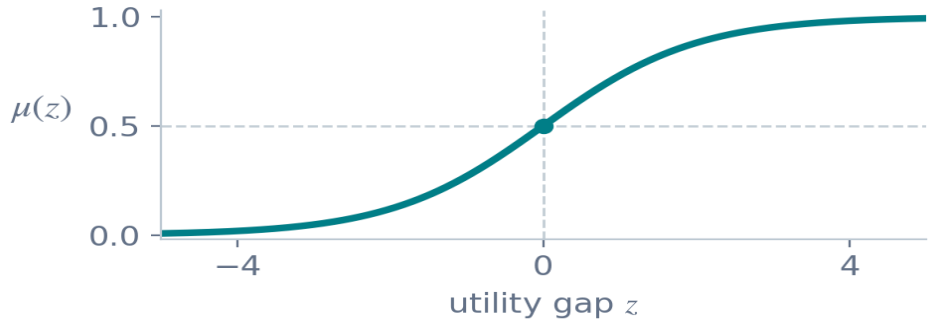
Preference probabilities and utility regret

Win probabilities describe the feedback; regret measures the quality of our choices.

LOGISTIC / BTL FEEDBACK

$$\Pr(y_t = 1 \mid \mathcal{F}_{t-1}) = \mu(\langle \theta^\star, \tilde{x}_t \rangle)$$

$$\mu(z) = \frac{1}{1 + e^{-z}}$$



Equal utility: 50% wins. Higher utility: a higher win probability.

OBJECTIVE: BOTH ARMS SHOULD BE GOOD

$$x_t^\star \in \arg\max_{x \in \mathcal{X}_t} f(x)$$

Regret = sum of the two utility losses

$$r_t = [f(x_t^\star) - f(x_{t,1})] + [f(x_t^\star) - f(x_{t,2})]$$

$$R_T = \sum_{t=1}^{T} r_t$$

Source: Section 1, p.1; BTL model and regret definition

Bounded features and parameters control the scale

These bounds control curvature, design-matrix growth and ridge bias.

SCALE BOUNDS

$$\|x\|_2 \le 1$$

Bounded features

$$\|\theta^\star\|_2 \le S$$

Known radius S

Difference features and true scores

$$\|\tilde{x}_t\|_2 \le \|x_{t,1}\|_2 + \|x_{t,2}\|_2 \le 2$$

$$|\langle \theta^\star, \tilde{x}_t \rangle| \le 2S$$

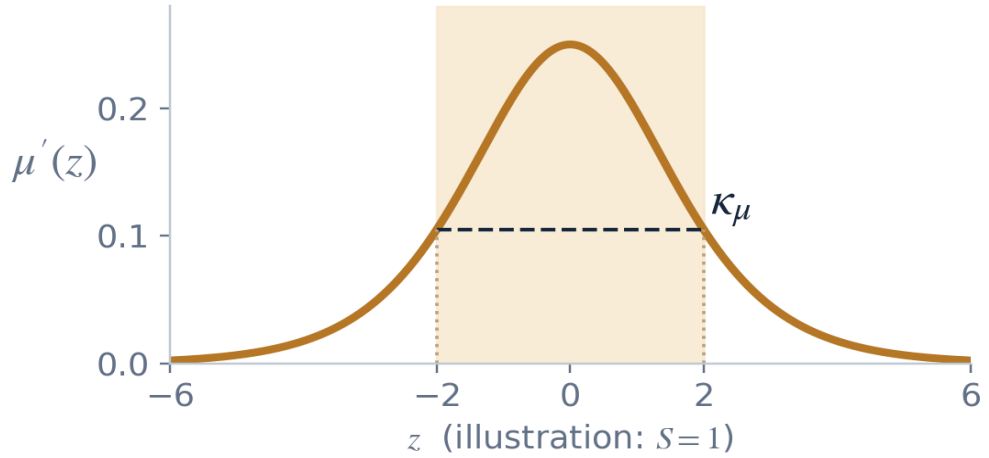
Role

Bounded utility gaps provide a scale for the estimation error.

Source: Section 1, Eqs. (1)(2); Section 2, Assumption 1; pp.1–2

Local curvature makes parameter error controllable

The logistic curve flattens in the tails; the proof needs curvature on the relevant scores.



ASSUMPTION 2

$$\mu'(z) = \mu(z)(1-\mu(z)) \geq \kappa_\mu > 0$$

Intermediate scores used in the proof:

$\xi_{s,t}$ between $\langle \theta^\star, \tilde{x}_s \rangle$ and $\langle \hat{\theta}_t, \tilde{x}_s \rangle$

Scope

A bound on the true parameter alone does not control the fitted scores.

The confidence result also needs curvature at every intermediate score.

Source: Section 2, Assumption 2; Section 4, Lemma 1, Step 2; pp.2, 5

Estimate utility with ridge logistic regression

Use the first t−1 rounds to estimate the parameter for round t.

1 Negative log-likelihood of one duel

Score $z = \langle \theta, \tilde{x}_s \rangle$

$$\ell(y, z) = -y \log \mu(z) - (1 - y) \log(1 - \mu(z))$$

2 Sum the losses and add ridge

$$\hat{\theta}_t = \arg\min_{\theta \in \mathbb{R}^d} \left\{ \sum_{s<t} \ell(y_s, \langle \theta, \tilde{x}_s \rangle) + \frac{\lambda}{2} \|\theta\|_2^2 \right\}$$

λ > 0 Strict convexity and stable estimation, with a ridge-bias term.

Source: Section 3.1, pp.2–4; Eqs. (4)–(8)

The first-order condition starts the proof

The logistic loss derivative simplifies to predicted probability minus observed label.

Score derivative

$$\frac{\partial \ell(y,z)}{\partial z} = \mu(z) - y$$

Parameter gradient

$$\nabla_\theta \ell(y_s, \langle \theta, \tilde{x}_s \rangle) = [\mu(\langle \theta, \tilde{x}_s \rangle) - y_s]\tilde{x}_s$$

At the minimizer

$$\sum_{s<t} [\mu(\langle \hat{\theta}_t, \tilde{x}_s \rangle) - y_s]\tilde{x}_s + \lambda \hat{\theta}_t = 0$$

Source: Section 3.2, p.4; Eqs. (9)(10)

Choose a greedy arm, then an optimistic challenger

Ridge-MLE Dueling UCB: exploit the current estimate, then explore through uncertainty.

First arm: highest estimated utility

$$x_{t,1} \in \arg\max_{x \in \mathcal{X}_t} \langle \hat{\theta}_t, x \rangle$$

Second arm: largest UCB relative to the first

$$x_{t,2} \in \arg\max_{x \in \mathcal{X}_t} \{ \langle \hat{\theta}_t, x - x_{t,1} \rangle + b \, \| x - x_{t,1} \|_{V_{t-1}^{-1}} \}$$

Design matrix

$$V_0 = \frac{\lambda}{\kappa_\mu} I_d$$

Update after the duel

$$V_t = V_{t-1} + \tilde{x}_t \tilde{x}_t^\top$$

The confidence bound determines b.

Source: Section 3, Algorithm 1, p.2; b = beta_T / kappa_mu

One confidence event covers every round

The target is a parameter-error bound; its radius comes from noise and regularization.

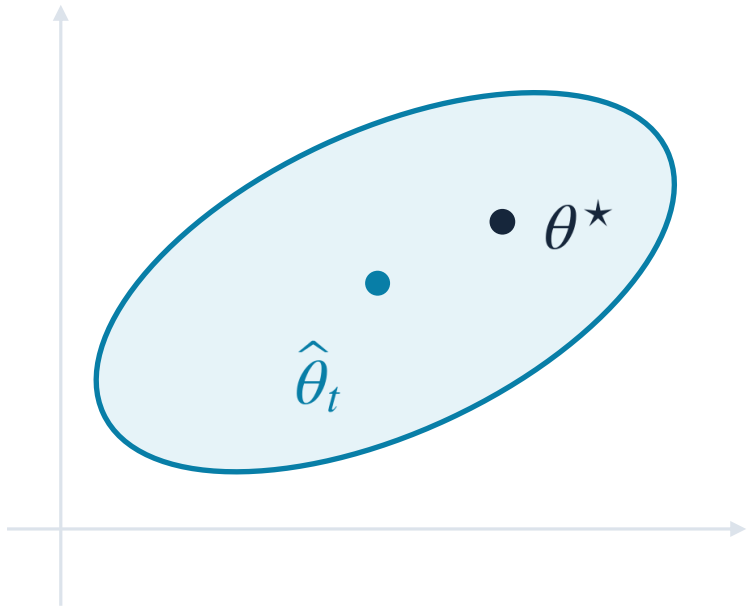
With probability ≥ 1−δ, for every t ≤ T:

$$\|\hat{\theta}_t - \theta^\star\|_{V_{t-1}} \le \frac{\beta_T}{\kappa_\mu} = b$$

$$\|u\|_V = \sqrt{u^\top V u}$$

V sets the shape; b sets the scale.

δ is the failure probability for all rounds.



Confidence region in parameter space (schematic)

Source: Section 4, Lemma 1, p.5; Eqs. (12)(13)

Proof sketch: noise plus regularization bias

$$\Delta_t = \widehat{\theta}_t - \theta^\star \qquad \varepsilon_s = y_s - \mu(\langle \theta^\star, \tilde{x}_s \rangle), \quad \mathbb{E}[\varepsilon_s \mid \mathcal{F}_{s-1}] = 0, \quad |\varepsilon_s| \le 1$$

1 Stationarity + mean value theorem

$$H_t \Delta_t = N_t - \lambda \theta^\star \qquad N_t := \sum_{s<t} \varepsilon_s \tilde{x}_s$$

2 Curvature lower bound

$$H_t = \sum_{s<t} \mu'(\xi_{s,t}) \tilde{x}_s \tilde{x}_s^\top + \lambda I_d \succcurlyeq \kappa_\mu V_{t-1}$$

3 Cauchy–Schwarz + cancel the error norm

$$\|\Delta_t\|_{V_{t-1}} \le \frac{1}{\kappa_\mu} \left(\|N_t\|_{V_{t-1}^{-1}} + \sqrt{\lambda \kappa_\mu}\, S \right)$$

Source: Section 4, Lemma 1, pp.5–6; Eqs. (14)(15)

The confidence bound determines the UCB bonus

Self-normalized concentration controls adaptive noise; Cauchy–Schwarz gives a utility-gap bound.

Radius numerator = ridge bias + noise bound

$$\beta_T = \sqrt{\lambda \kappa_\mu}\, S + \sqrt{2\log(1/\delta) + d\log\left(1 + \frac{4\kappa_\mu T}{\lambda d}\right)}, \qquad b = \frac{\beta_T}{\kappa_\mu}$$

Simultaneously for every pair of arms x, x′:

$$|f(x) - f(x') - \langle \hat{\theta}_t, x - x' \rangle| \le b \, \|x - x'\|_{V_{t-1}^{-1}}$$

Hence Estimated utility gap + confidence bonus = the UCB for the second arm.

Source: Section 4, Lemma 1, Step 5 and Corollary 1, pp.6–7; Eqs. (12)(16)(17)

Connecting the four parts

From a preference observation to a comparison backed by a confidence guarantee.

01

Problem

Compare two arms; observe a preference.

→

02

Assumptions

Boundedness; local curvature.

→

03

Algorithm

Ridge-MLE estimate; greedy arm + UCB.

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04

Confidence

Parameter bounds; utility-gap bounds.

Confidence defines the exploration bonus, which guides the next comparison.